

MSI's 'Big Bang' is the first motherboard model to utilize the Lucid HYDRA chip which allows to use NVIDIA and ATI based cards for multi-GPU scaling



Motherboard and GPU maker ZOTAC launches its first NVIDIA ION based Nettop with a dual-core Intel Atom, 2 GB memory and 160GB of HDD space

Motherboard Test

HOW WE TESTED

For all seven boards, we used Intel's Core i7 870 processor running at 2.93 GHz, since this is the fastest processor till date for the P55 platform. Corsair's Dominator dual-channel DDR3 4GB (1333 MHz) kit was used. During the test, Windows Vista Ultimate Edition ran off a Western Digital 300GB

Velociraptor HDD. Graphics was provided by an Asus GeForce 9800GTX+. All the boards were powered by Corsair's HX620 PSU. We ran PCMark Vantage to test the performance across various usage scenarios. That was followed by the ScienceMark Memory Bandwidth test. CineBench R10, an independent tool based on

the popular 'Cinema 4D' 3D modelling program was used to test just the processor and memory. Another way to test the processor was by using WinRAR 3.9s built-in benchmark tool.

Coming to the real world tests, we tested the time taken to encode a .VOB video file into the DivX format using DivX Encode 7.0. That was

followed by a single large file and assorted 8GB file transfer test to simulate sequential and assorted from one partition to the other.

In order to see how these boards performed in the graphics department, we tried running S.T.A.L.K.E.R Clear Sky benchmark and the Crysis CPU and GPU benchmarks as well.

processor directly; thus reducing the CPU-GPU data transmission latency and bottlenecking that occurs as a result. This brings us to another shortcoming of the P55 chipset as compared to the higher X58 chipset that might displease gamers who want to play every game at its best quality. The P55 will support two graphics cards that will utilise those 16 lanes of PCI-E 2.0 only at x8 mode, whereas the X48 supports up to 36 lanes, hence supporting two graphics cards running at full x16 speed.

Since Core i5 is currently the hot stuff, we decided to give you a comparison of the latest motherboards based on the P55 platform.

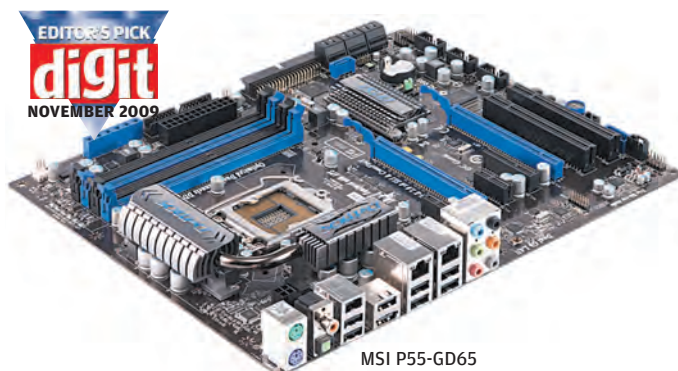
Biostar TPOWER I55

The first thing that the Biostar TPOWER I55 reminded us of was *The Transformers*. This resemblance we feel is pertaining to the three heat-sink units surrounding the CPU – a very Optimus Prime-ish look. Overall, the quality of the materials used seems to be pretty good. All that metal makes the board feel heavy and pretty sturdy. The heat pipes are also built well. There are dedicated power/reset switches

and a two-digit LED status code display. All six SATA ports are placed in an array on the right corner. Nowadays, motherboard manufacturers are skipping the keyboard/mouse PS/2 ports altogether. The Biostar has at least had the decency to place the PS/2 keyboard port on the board. Automatic or manual over-clocking can be performed either through the BIOS or via a Windows-based application.

MSI P55-GD65

The MSI board has two heatsinks as compared to three on the Biostar board above. Now this frees up a little space and possibly makes space for one more PCIe x1 slot. The build of the board seems to be quite satisfactory. MSI is trying to win overclockers' hearts by providing hardware buttons on the board itself. There's one that activates the automatic overclocking feature and a plus/minus button to change the CPU base-clock without any kind of software/BIOS intervention. MSI also provides a couple of utilities bundled with the package. Some mentionable ones are a light-weight fast-booting Linux distribution called Winki.



MSI P55-GD65

Unlike quick-boot OSES we've seen earlier that are directly embedded onto a separate memory on the motherboard, Winki runs via either a bootable USB drive or DVD. Next is a nifty bootable HDD backup utility. Another cool feature on this board is the 'Power eSATA' port, which means that eSATA devices will not require an additional power cable while connecting to the eSATA port.

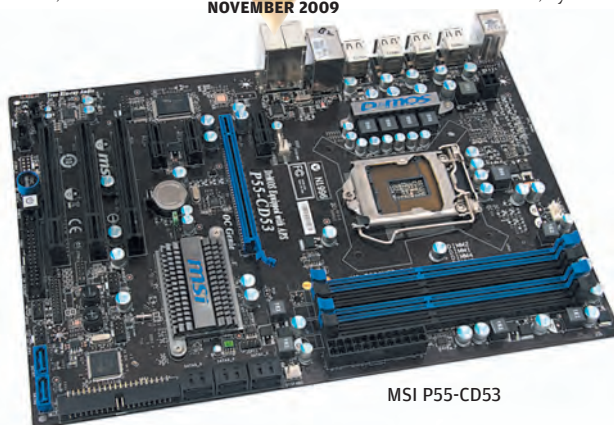
feels pretty spaced-out since it has one tiny heatsink as opposed to the dual-piped heat-sinks on the GD65. The RAM slots are packed closely. The CD53 bears a single PCI Express slot, and thus isn't meant for a SLI/CrossFire setup. The board's build otherwise is quite similar to the GD65 and is satisfactory. The auto overclocking and base clock tweaking buttons are still present.

MSI P55-CD53

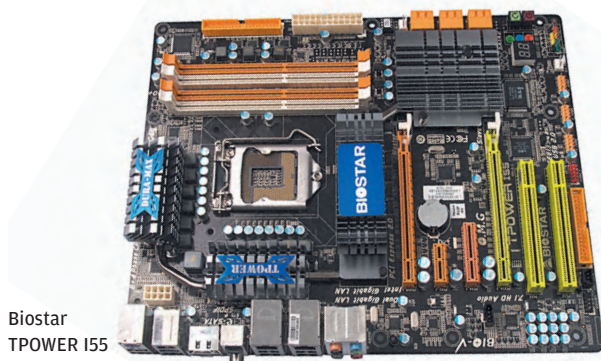
Hot on the heels of its older brother – the GD65, comes the P55-CD53. To cite the major differences, this board

ASUS P7P55D Deluxe

We'd call the P7P55D from ASUS as something that isn't commonly associated with motherboards; eye



MSI P55-CD53



Biostar TPOWER I55